Sudarshan Babu

Research Scientist – Generative Modeling

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Research Summary

Research scientist developing generative models for data-scarce domains. Specializes in 3D generation, including scenes, small molecules, cryo-ET volumes, and single-cell modeling using few-shot learning methods, hypernetworks, and meta-learning.

Core Expertise

Research — Diffusion models, 3D generation, small molecule design, single cell modelling

Methods — Hypernetworks, implicit neural representations (INRs), transformers, meta-learning

Professional Experience

Sep 2025–Present — **Research Scientist**, *Chan Zuckerberg Biohub Chicago*, Chicago, IL.
- Developed scLDM, a latent diffusion model for single-cell data generation
- Built implicit neural field model for cryo-ET volume generation

Jan 2025–Sep 2025 — **AI Fellow**, *Chan Zuckerberg Biohub Chicago*, Chicago, IL.
- Developed HyperDiffusionFields (HyDiF), combining diffusion models with hypernetworks for implicit molecular neural fields generation

Oct 2021–Jan 2022 — **Research Scientist Intern**, *Amazon*, Bellevue, WA..
- Designed curriculum learning strategies for efficient neural network training

Feb 2021–Jun 2021 — **Research Scientist Intern**, *NVIDIA*, Remote.
- Built systems for training models on non-stationary long-tail distributions
- Contributed to 4 patent applications in adaptive learning and distributed training systems

Education

Oct 2019–2025 — **Ph.D., Computer Science**, *Toyota Technological Institute at University of Chicago*, Chicago, IL, GPA: 3.90/4.0.
- Advisors: Prof. Michael Maire, Prof. Greg Shakhnarovich
- Dissertation: **Acquiring and Adapting Priors for Novel Tasks via Neural Meta-Architectures**

Selected Papers

ICML '26 — **Scalable Latent Diffusion Model for Single-Cell Gene Expression Data**
*Giovanni Palla, **Sudarshan Babu**, et al.*

ICML '26 — **Virtual Cells Need Context, Not Just Scale**
*Payam Dibaeinia, **Sudarshan Babu**, et al.*

NeurIPS WS '25 — **HyperDiffusionFields (HyDiF): Diffusion-Guided Hypernetworks for Learning Implicit Molecular Neural Fields**
*__Sudarshan Babu*__, Phillip Lo*, et al.*

ICML '24 — **HyperFields: Towards Zero-Shot Generation of NeRFs from Text**
*__Sudarshan Babu*__, Richard Liu*, Avery Zhou*, et al.*

CVPR '21 — **Domain-Independent Dominance of Adaptive Methods**
*Pedro Savarese, David McAllester, **Sudarshan Babu**, Michael Maire*

NeurIPS '21 — **Online Meta-Learning via Learning with Layer-Distributed Memory**
__Sudarshan Babu__, Pedro Savarese, Michael Maire

Preprint — **HyperNetwork Designs for Improved Classification and Robust Meta-Learning**
__Sudarshan Babu__, Pedro Savarese, Michael Maire

Service

Reviewing — CVPR, ICCV, ECCV, NeurIPS, ICLR, SIGGRAPH, RECOMB

Teaching — Teaching Assistant for Machine Learning, Prof. Greg Shakhnarovich, Fall 2020

Organization — Co-organizer: Computational Biology Workshop @ TTIC, Sep 2024; Annual TTI-C Student Workshop, Feb 2020; Computer Vision Reading Group, Oct 2020 – Feb 2024